AI WEATHERWISE

Deliver intelligent weather forecasts and insights.

PROJECT DOCUMENTATION

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1. **ABSTRACT**

AI WeatherWise is an intelligent weather information system that provides real-time weather data, AI-powered insights and weather-based recommendations.

The system integrates OpenWeatherMap API with Google Gemini AI to convert weather data into simple and meaningful information.

2.INTRODUCTION

Weather information is important for daily activities, travel and outdoor planning.I WeatherWise combines real-time weather data and Artificial Intelligence to provide understandable weather insights and useful recommendations through RESTful APIs.

3.PROBLEM STATEMENT

Traditional weather applications mainly provide raw weather information.

Major Problems:

- Difficult to interpret raw data
- Limited AI-based insights
- Limited weather recommendations
- Information may not be presented in a simple form

4.OBJECTIVES

- Provide real-time weather information
- Generate AI-powered weather insights
- Provide weather-based recommendations
- Implement secure user authentication
- Develop RESTful APIs
- Store user data securely

5.PROPOSED SYSTEM

AI WeatherWise integrates Weather API, REST API and Artificial Intelligence into one system.

Key Features:

• Real-Time Weather

• AI Weather Insights

• Smart Recommendations

• JWT Authentication

• MongoDB Database

• RESTful APIs

6.TECHNOLOGY STACK

Backend:

Node.js, Express.js

Database:

MongoDB, Mongoose

Authentication:

JWT, bcryptjs

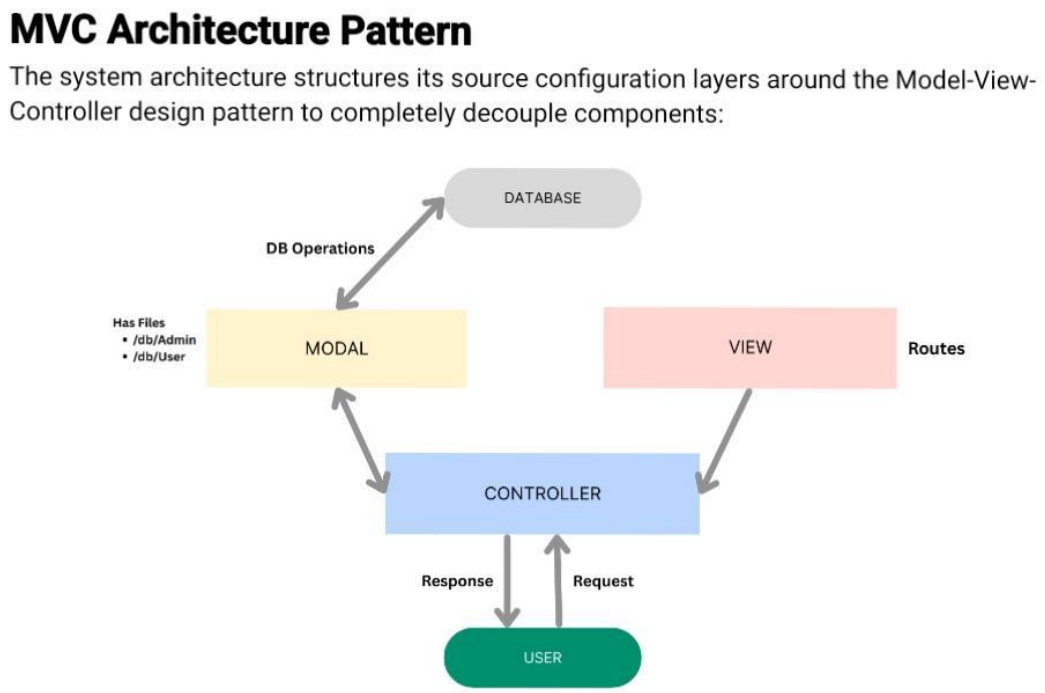
APIs:

OpenWeatherMap API, Google Gemini AI

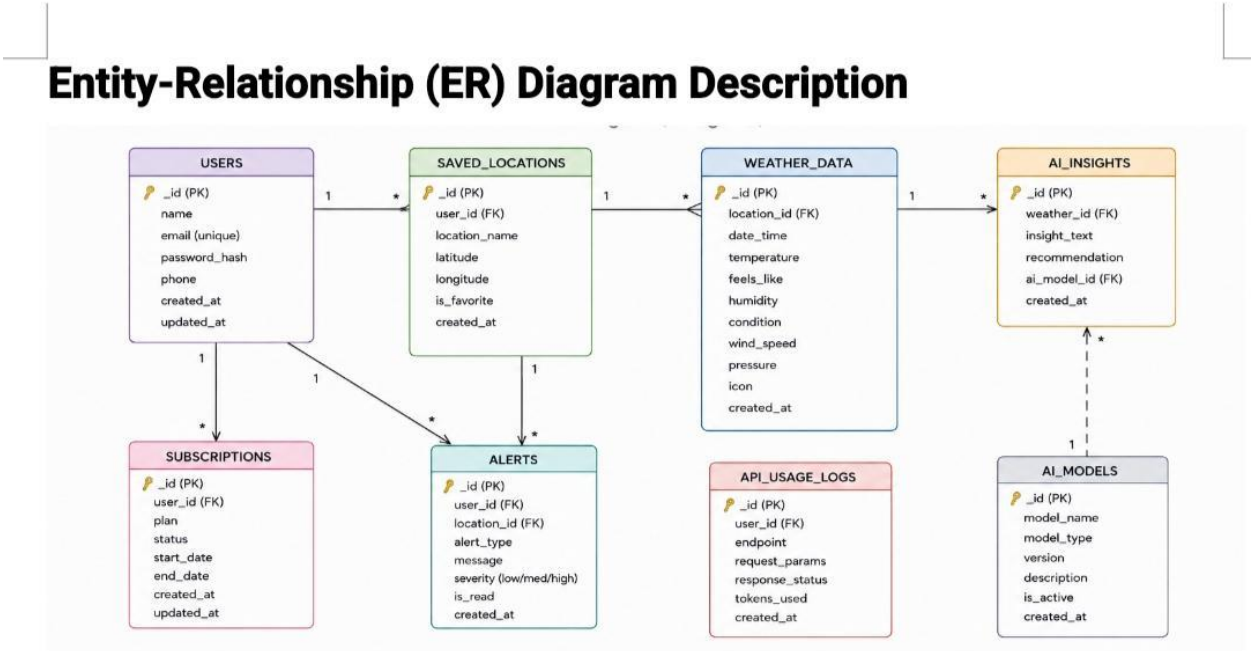
Testing:

Thunder Client

7.SYSTEM ARCHITECTURE



ENTITY-RELATIONSHIP (ER) DIAGRAM



8.MODULES

1. User Authentication

Provides user registration, login and JWT-based authentication.

2. Weather Module

Retrieves real-time weather information using OpenWeatherMap.

3. AI Insights Module

Uses Gemini AI to analyse weather data and generate summaries.

4. Recommendation Module

Provides useful suggestions based on weather conditions.

5. Database Module

Stores user information using MongoDB.

2. USER AUTHENTICATION

Registration:

Creates a new user account using name, email and password.

Login:

Verifies user credentials and generates a JWT token.

Security:

• Password Hashing

• JWT Authentication

• Protected Routes

• Environment Variables

9.WEATHER API

The Weather API retrieves real-time weather information using the OpenWeatherMap API.

Sample Input:

{

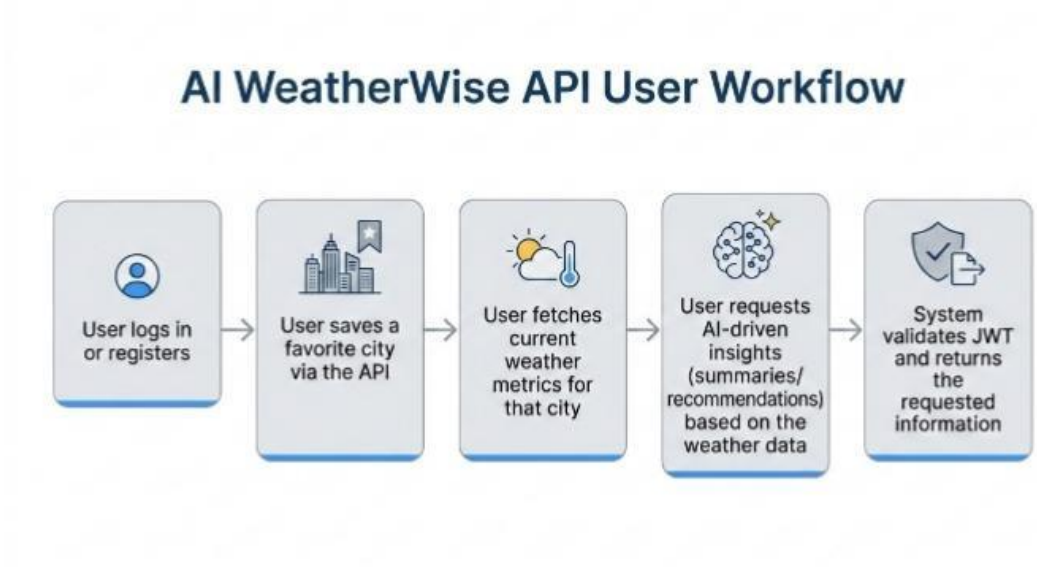
"city": "Chennai"

}

Weather Data:

- Temperature
- Humidity
- Wind Speed
- Weather Condition

WEATHER API PICTURE



10.RECOMMENDATION

The system provides recommendations based on weather conditions.

Examples:

Hot Weather:

Hydration advice

Rainy Weather:

Rain precautions

Strong Wind:

Outdoor safety

High Temperature:

Heat precautions

11.API TESTING

Thunder Client is used to test the APIs.

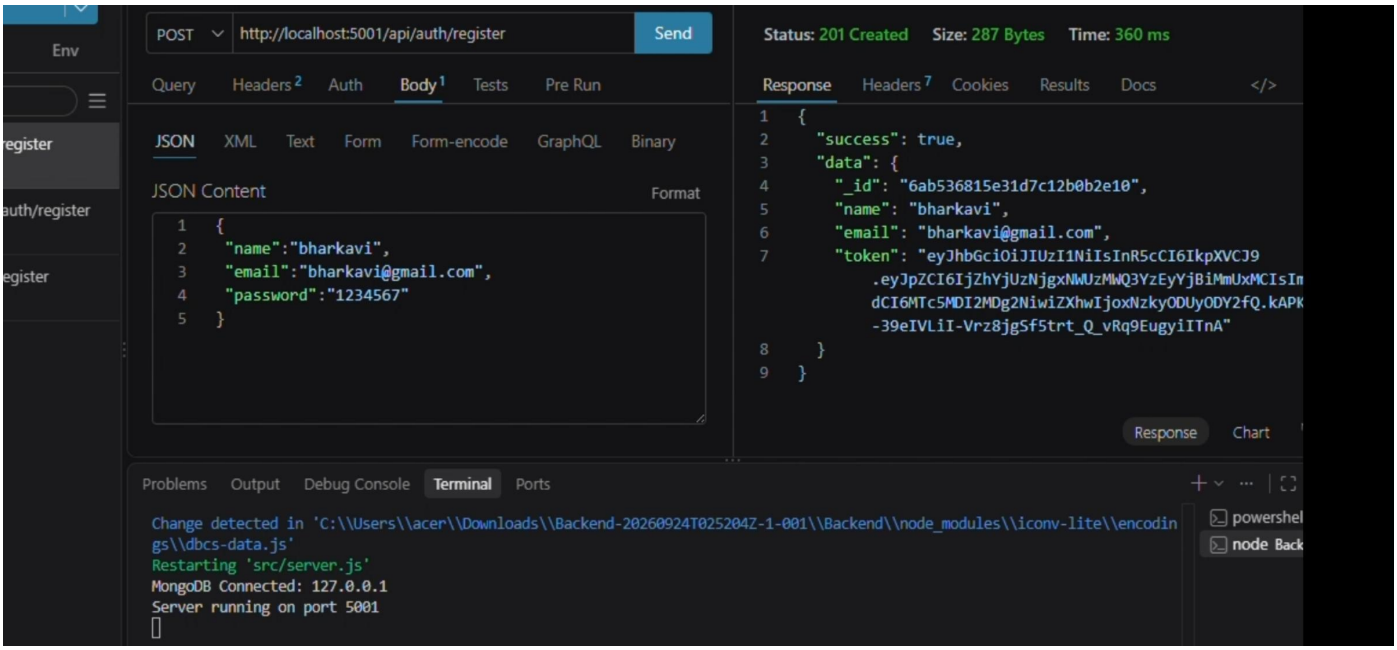
Tested APIs:

1. Register
2. Login
3. Weather
4. AI Summary
5. Recommendation

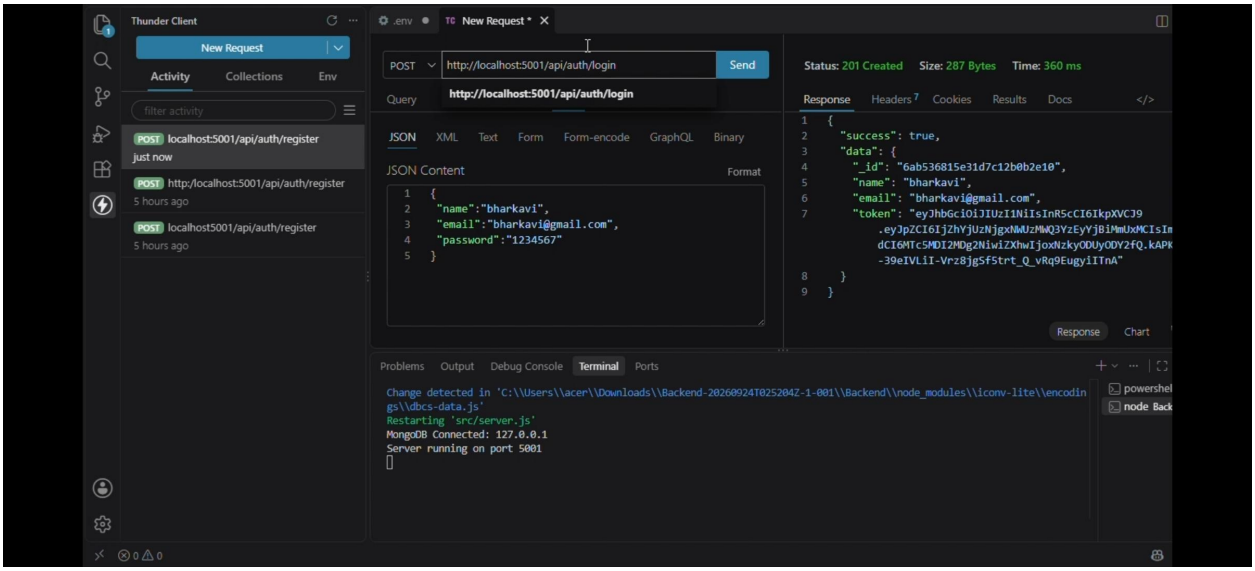
Testing Flow: Request → Server → Processing → Response

12.API SCREENSHOT PLACEHOLDERS

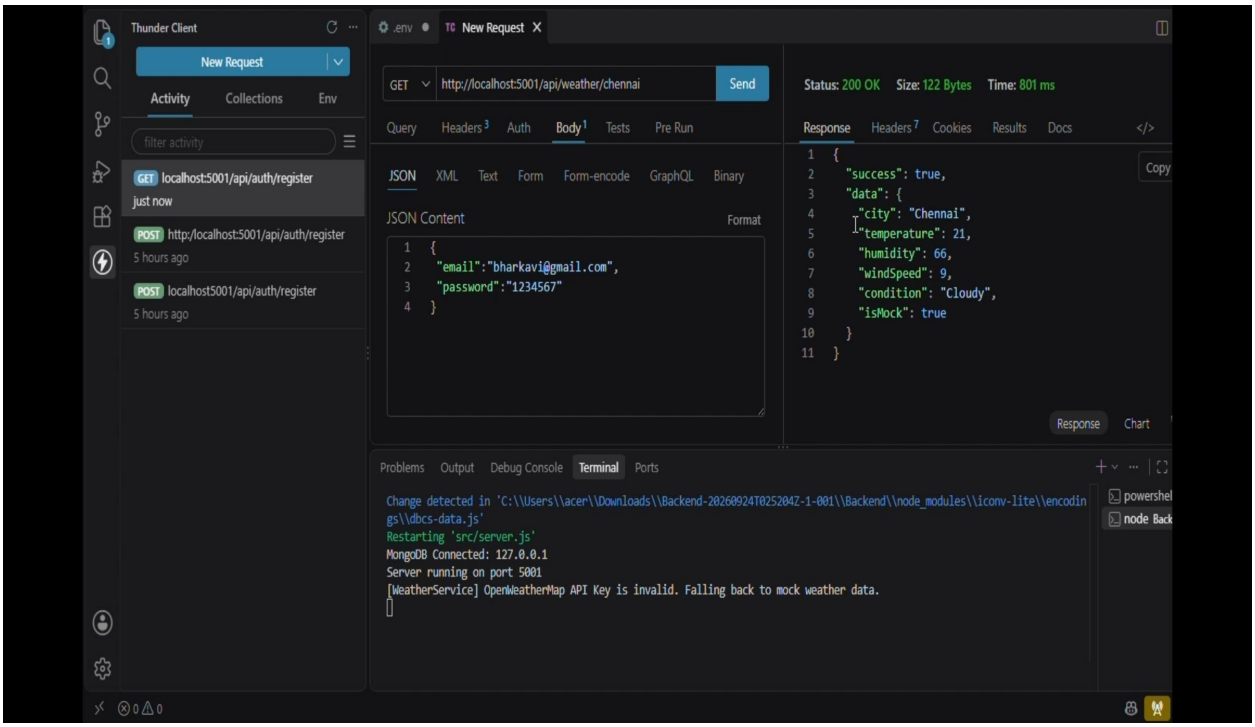
Register API



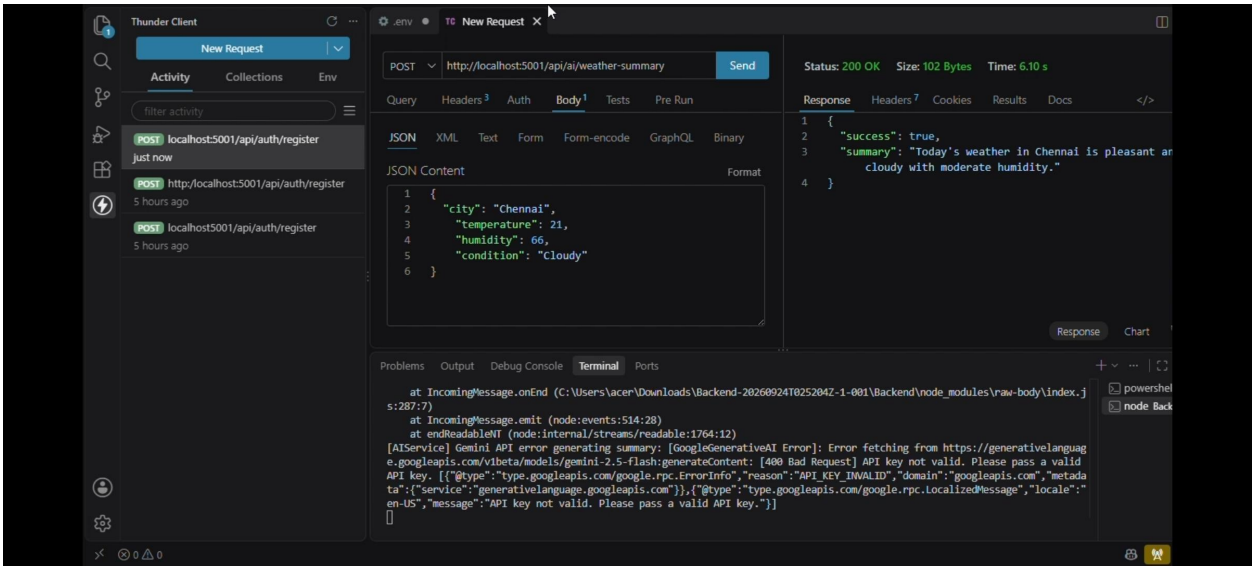
Login API



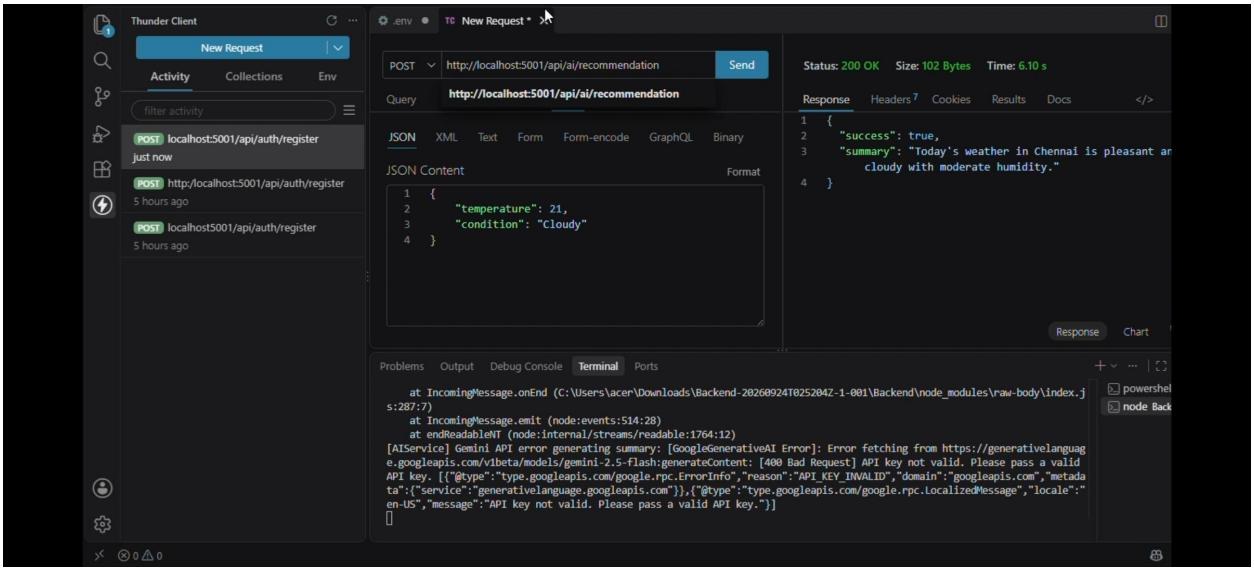
Weather API



Weather Summary



Recommendation API



13.ADVANTAGES AND FUTURE ENHANCEMENTS

Advantages:

- Real-time weather information
- AI-powered insights
- Smart recommendations
- Secure authentication
- Scalable REST API

Future Enhancements:

- Mobile Application
- Weather Alerts
- Voice Assistant
- Weather Analytics
- Multi-Language Support
- Advanced AI Recommendations

14.RESULT AND CONCLUSION

Result:

AI WeatherWise successfully provides real-time weather information, AI-powered insights and weather-based recommendations through RESTful APIs.

Conclusion:

The project demonstrates the integration of Node.js, Express.js, MongoDB, OpenWeatherMap and Google Gemini AI to create an intelligent weather information system.

DEMO VIDEO :
https://drive.google.com/file/d/1ZJfVSXeOc3RlszLp13jzF2Pi1xa6UQlQ/view?usp=drivesdk

THANK YOU